



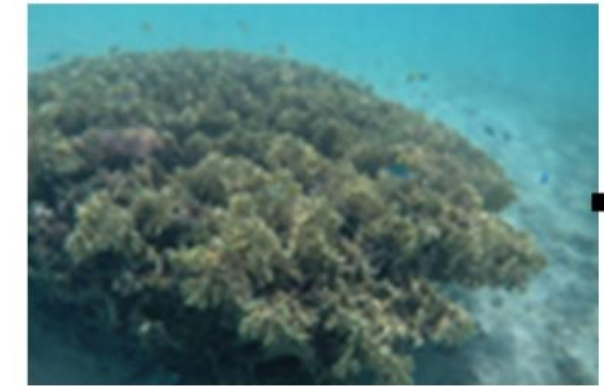
WAVELIFT: FAST FREQUENCY-GUIDED UNDERWATER RESTORATION

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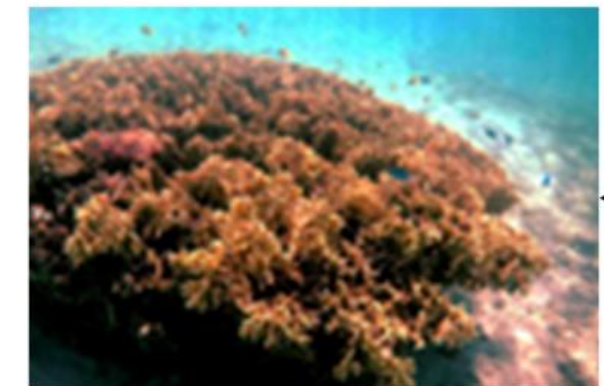


Application Domain: Underwater Image Enhancement

- Underwater images often look too blue/green, low-contrast
- This matters for underwater robots, marine inspection, real-time object detection, etc.
- The goal is to restore realistic colors and sharp details from degraded underwater photos.



Input

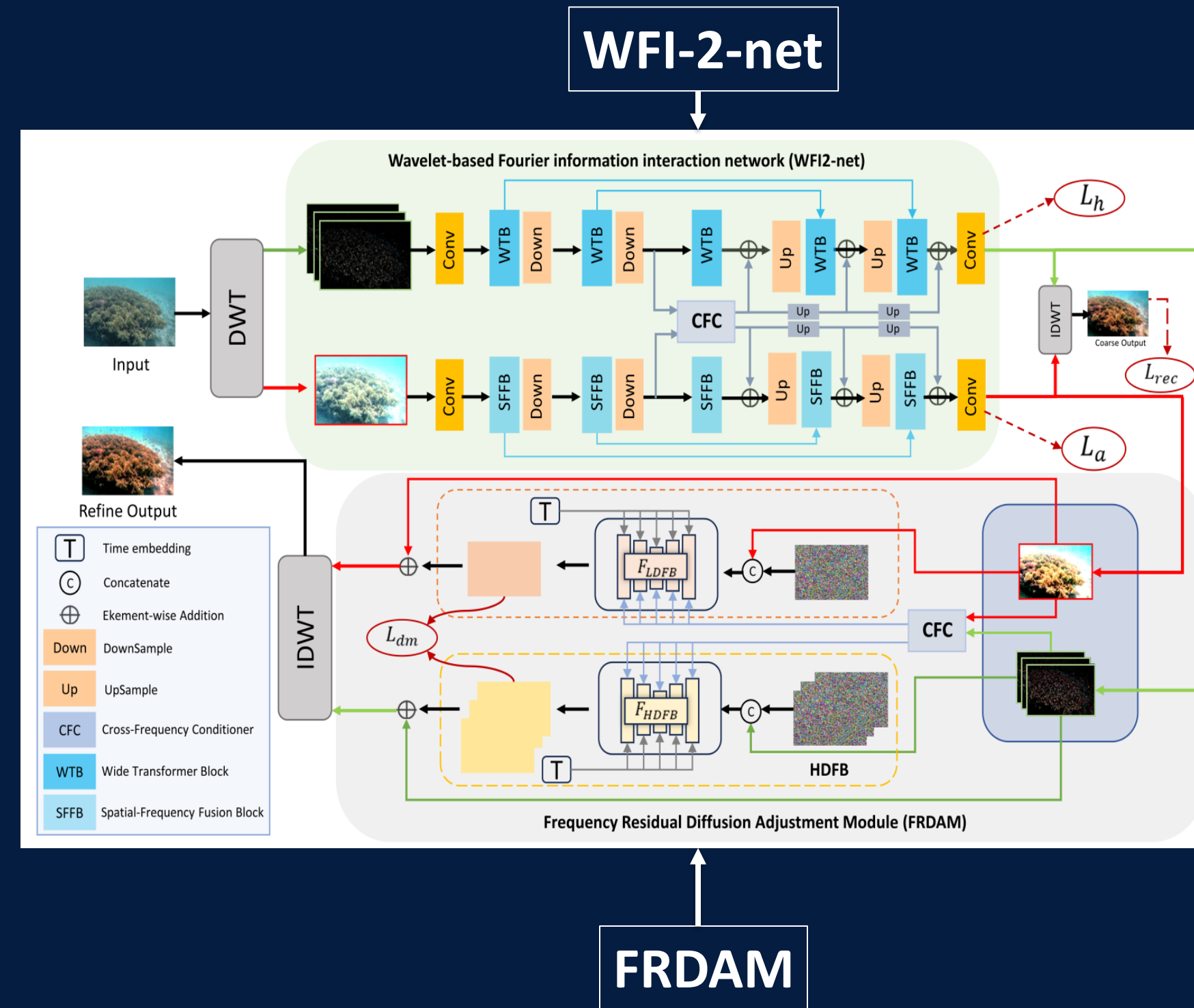


Refine Output

Adapted from Zhao *et al.* (CVPR 2024)

Base Model and Datasets

- Reference Model: **WF-Diff** with 2 stages/modules
 - WFI2-net** that does the main improvement
 - FRDAM** has 2 branches deals with high frequency (details/edges) and low-frequency (color/illumination) separately.
- UIEB Dataset: 890 paired images (raw underwater + a “reference” improved image) and 90 non-paired images.
- UIEB Dataset: 5004 paired images, much more diverse in underwater environments

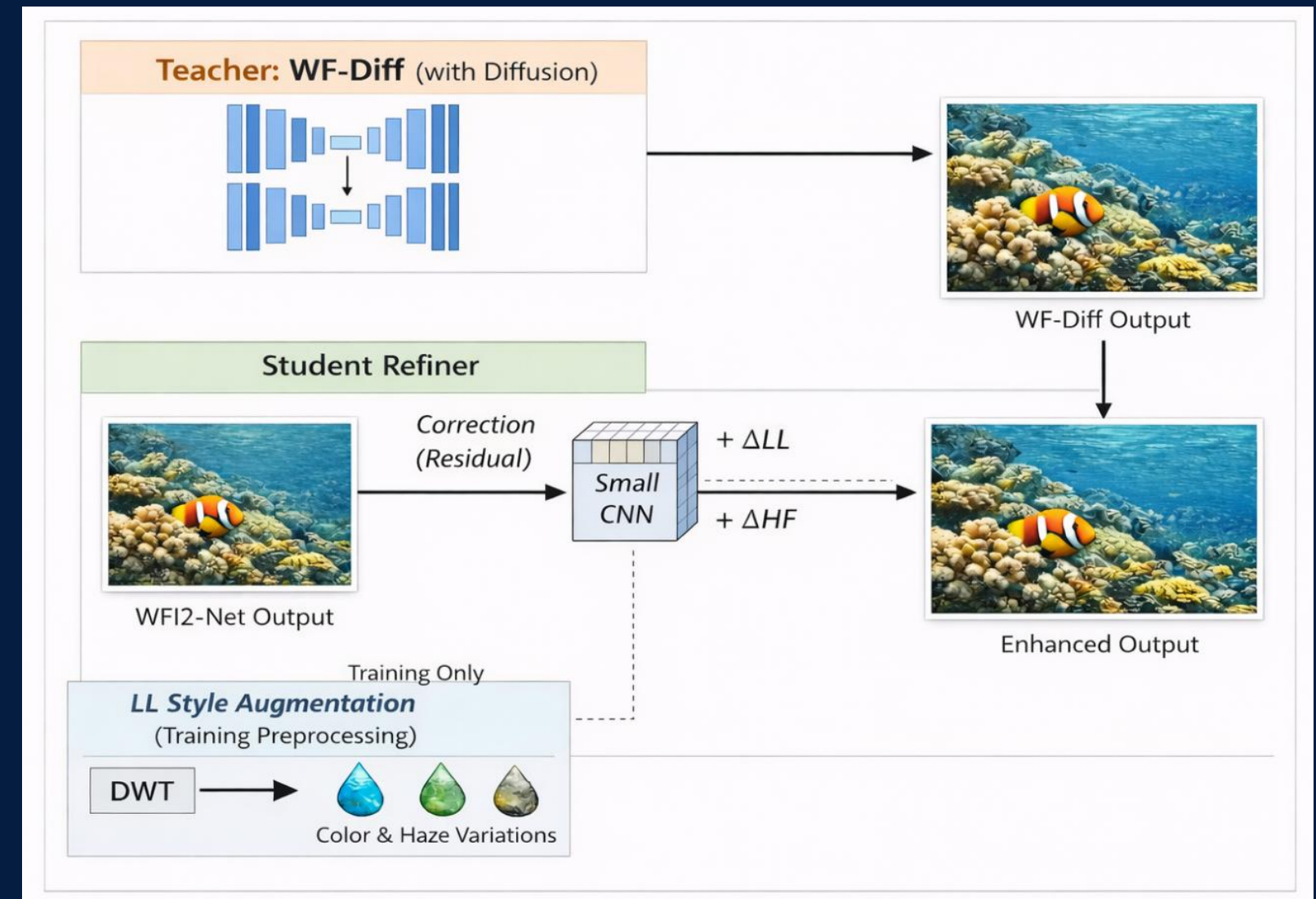


Current Problems with WF-Diff

- WF-Diff can look great on the training dataset, but performance drops on new environments/pictures.
 - Sensitive to water type and/or camera changes
- FRDAM uses 2 different diffusion branches, which affects inference speed massively
- Goal: make enhancement stable across different underwater styles and to lower the inference time.

Proposed Improvements

- Use full WF-Diff (with diffusion) as the teacher to generate high-quality outputs, then train a small student refiner (2 CNNs) to mimic the teacher's refinement.
- The CNNs predict a small correction (residual) that is added to WFI2-net's output to match teacher quality.
- Apply pre-process low-frequency style randomization (simulate different watercolors, haze, and lighting).



Rough Sketch for the Improvement Architecture